UNIT - II

Operations Management

Syllabus at a glance

- Plant layout, methods of production, work study, SQC, material management (EOQ, ABC)
- Marketing management: concept, functions, marketing mix, channels, advertisement and sales promotion, PLC strategies

A. OPERATIONS MANAGEMENT

1. Meaning & scope

Operations (Production) Management = planning, organising, directing and controlling the activities that convert inputs (men, material, machines, money) into outputs (goods and services) at the right quality, quantity, time and cost.

- Inputs → Transformation process → Outputs (with feedback and control)
- Scope: product design, plant location, plant layout, production planning and control, quality control, material handling, maintenance, work study
- Objectives: right quality, right quantity, right time, right cost (the 4 R's)

2. Plant Layout

Plant layout = arrangement of machines, equipment, materials, workers and service areas within the factory building so that production is smooth, economical and safe.

Objectives

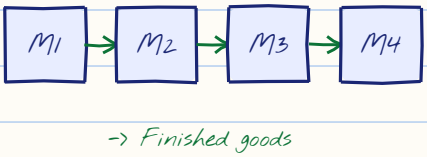
- Smooth flow of material and minimum handling - Better use of floor space - Reduce production cycle time
- Safety and comfort of workers - Flexibility for future changes - Reduce WIP inventory and cost

Principles of a good layout (Moore)

1. Integration - combine men, material, machine in one system
2. Minimum distance - shortest path for material movement
3. Flow principle - arrange in the order of operations, avoid back-tracking
4. Cubic space - use vertical space too (not only floor)
5. Satisfaction and safety - for workers
6. Flexibility - easy rearrangement

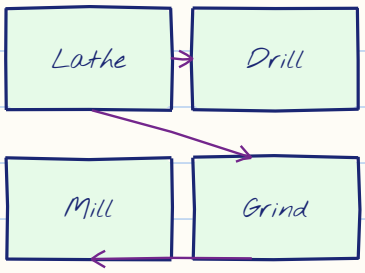
Types of layout

PRODUCT (line) layout



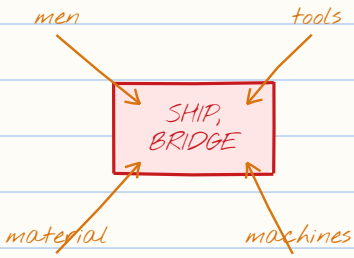
one product, fixed sequence

PROCESS (functional) layout



similar machines grouped, many products

FIXED-POSITION layout



product stays, resources move

Type	Merits	Demerits
Product (line)	Low handling cost, less WIP, simple control, fast output, less skilled labour	Costly, inflexible, one breakdown stops all line, monotony
Process (functional)	Flexible, general-purpose machines, easy to handle breakdown, skilled workers' growth	More handling, higher WIP, difficult control, long cycle time
Fixed position	No movement of heavy product, flexible, good for large products	Costly movement of men and tools, low machine use, space needed
Cellular (Group technology)	Combines product and process benefits, shorter set-up, high flexibility	Needs part families, more planning

Which layout where?

- Product layout → mass production (cars, bottling, sugar) - Process layout → job or batch (machine shops, hospitals)
- Fixed position → ships, aircraft, bridges, dams - Group layout → families of similar parts (cells)

3. Methods (Types) of Production

Basis	Job production	Batch production	Mass (flow) production
Meaning	One or few units as per customer order	Limited quantity produced in lots or batches	Large standardised volume, continuous
Product	Non-standard, custom-made	Semi-standard	Highly standard
Machines	General purpose	General purpose + some special	Special purpose, automatic
Layout	Process or fixed position	Process or group	Product layout
Labour	Highly skilled	Semi-skilled	Low skilled
Cost per unit	Highest	Medium	Lowest
Flexibility	Maximum	Moderate	Minimum
Examples	Ships, bridges, tailor-made suits, dams	Bakery, medicines, books, shoes	Cars, cement, sugar, refrigerators, bulbs

- Continuous (process) production : non-stop flow, e.g. oil refining, chemicals, fertilizers - also a type of mass production
- Intermittent production = job + batch ; Continuous = mass + process

4. Work Study

Work study = systematic examination of methods of carrying out activities so as to improve effective use of resources and set standards of performance (ILO).

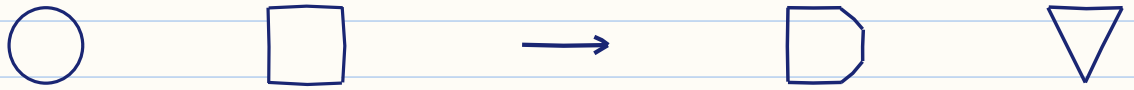
Two parts: Method study + Work measurement. Aim: higher productivity with no extra capital.

(A) Method Study - "Find the best way"

Steps - SREDDIM

1. Select the job to be studied 2. Record all facts (charts, diagrams) 3. Examine critically (what? why? where? when? who? how?)
4. Develop the best method 5. Define the new method 6. Install it 7. Maintain it

- Recording tools: Outline process chart, Flow process chart, Flow diagram, Man-machine chart, Multiple-activity chart, SIMO chart
- Critical examination - ECRS: Eliminate, Combine, Rearrange, Simplify



Operation Inspection Transport Delay Storage

(B) Work Measurement (Time study)

Work measurement = determining time required by a qualified worker to do a task at a defined level of performance. Techniques: stop-watch time study, work sampling, synthetic data (PMTS, MTM), standard data.

- Steps of time study: select job → record method → break into elements → time with stop-watch (several cycles) → rate the worker → add allowances → compute standard time

Formulae

- Normal time = Observed time × (Rating % ÷ 100)
- Standard time = Normal time + Allowances = Normal time × (1 + Allowance %)
- Allowances: personal needs, fatigue, delay, contingency (total about 10–20%)

Example

- Observed time = 2.0 min, performance rating = 110%, allowances = 15%
- Normal time = 2.0 × 1.10 = 2.20 min
- Standard time = 2.20 × 1.15 = 2.53 min per unit → in an 8-hour shift

(480 min) expected output = 480 ÷ 2.53 = ~190 units

5. Statistical Quality Control (SQC)

Quality = fitness for use or conformance to specification. SQC = use of statistical techniques (sampling, control charts) to monitor and control the quality of a process or product. Pioneer: Walter A. Shewhart (1924, Bell Labs).

Causes of variation

- Chance causes (random, natural) - small, unavoidable, process is "in control" e.g. slight temperature changes
- Assignable causes (special) - due to tool wear, wrong setting, bad material, operator error - can be identified and removed → process "out of control"

Two major tools

- Acceptance sampling : inspect a sample from a lot and accept or reject the lot. Risks: producer's risk (alpha, good lot rejected), consumer's risk (beta, bad lot accepted). Terms: AQL, LTPD, OC curve, sampling plan (single, double, multiple)
- Process control (control charts) : plot sample measure against time with 3-sigma limits (UCL, CL, LCL)

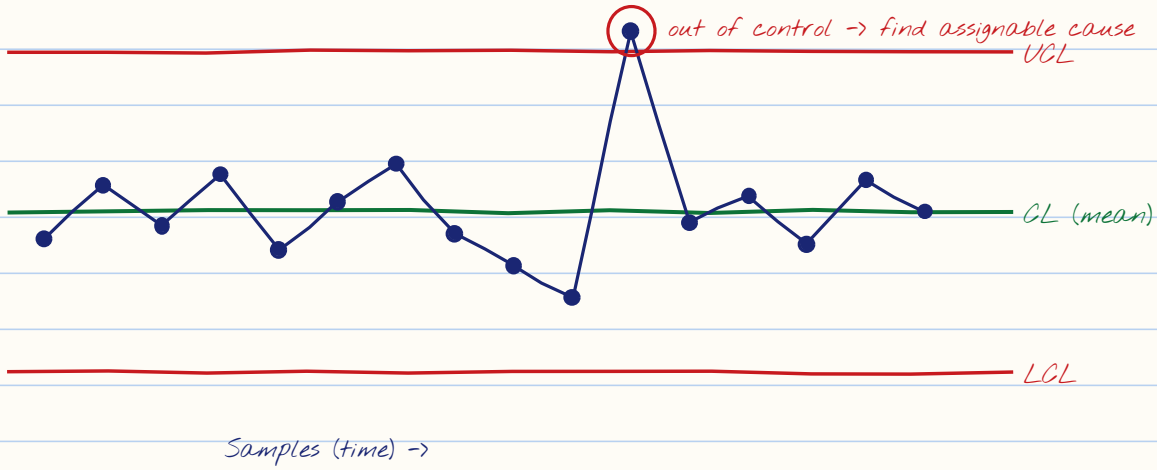


Chart	Type	Used for
X-bar and R chart	Variables	Mean and range of measurable data (length, weight)
p chart	Attributes	Fraction defective (variable sample size)
np chart	Attributes	Number of defectives (fixed sample size)

Chart	Type	Used for
c chart	Attributes	Number of defects per unit (cloth flaws, scratches)

Control limit formulae

- X-bar chart : UCL = X-bar + A2 × R-bar , LCL = X-bar − A2 × R-bar (n = 5 → A2 = 0.577)
- R chart : UCL = D4 × R-bar , LCL = D3 × R-bar (n = 5 → D4 = 2.114, D3 = 0)
- p chart : p-bar ± 3 × sqrt(p-bar (1 − p-bar) ÷ n) ; c chart : c-bar ± 3 × sqrt(c-bar)

Example

- X-bar = 50 mm, R-bar = 4 mm, n = 5 : UCL = 50 + 0.577×4 = 52.31 , LCL = 50 − 2.31 = 47.69 ; R chart UCL = 2.114×4 = 8.46 , LCL = 0
- c chart, c-bar = 9 defects: UCL = 9 + 3×3 = 18 , LCL = 9 − 9 = 0

- Benefits of SQC: early detection of faults, less rework and scrap, lower inspection cost, better customer satisfaction, basis for TQM

6. Material Management

Materials management = planning, organising and controlling the flow of materials from identification of need to the point of delivery to production (purchasing, stores, inventory, handling, disposal). Materials form 50–60% of the cost of a product - so a small saving matters a lot!

Objectives

- Right material, right quantity, right quality, right place, right time, right price (the "6 R's")
- Minimum inventory and storage cost - Continuous supply to production - Good vendor relations - Standardisation and value analysis - Reduce wastage, obsolescence

Functions

- Material planning and budgeting - Purchasing (vendor selection, tenders, negotiation) - Receiving and inspection - Storing and stores accounting -

Inventory control - Material handling - Disposal of scrap and surplus

7. Inventory

Inventory = stock of materials held for future use or sale: raw materials, components, WIP, finished goods, spares, consumables.

Functions (why hold stock?)

- Ensures continuous production - Protects against uncertainty in supply or demand (buffer)
- Gets quantity discounts and saves ordering cost - Hedge against price rise or inflation - Smooths seasonal demand (anticipation stock) - Better customer service

Types of inventory

- Raw materials - Work-in-progress (WIP) - Finished goods - MRO (maintenance, repair, operating supplies) - Spares and tools
- By role: Cycle stock , Safety (buffer) stock , Anticipation stock , Pipeline (transit) stock , Decoupling stock

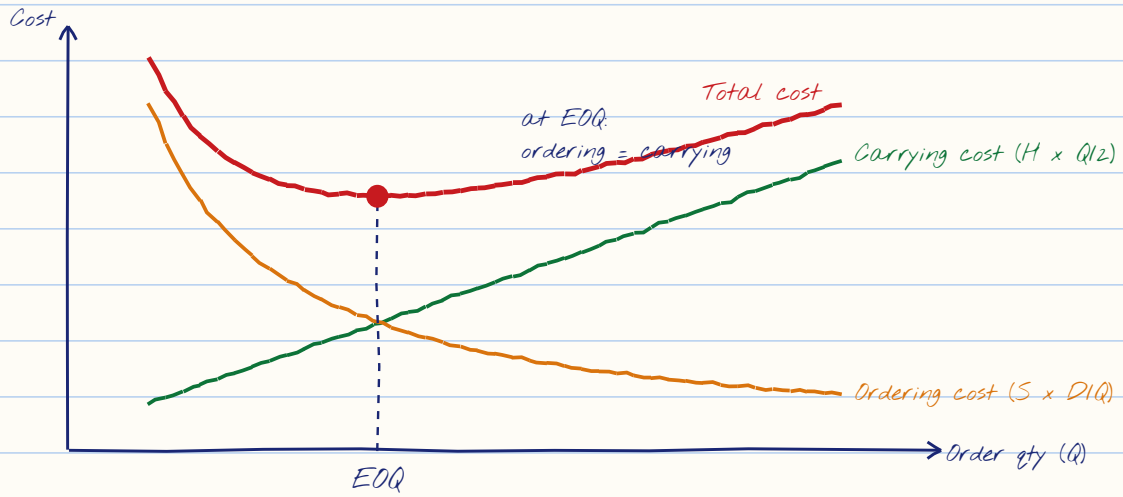
Inventory costs

- Ordering (procurement) cost - paperwork, transport, inspection (per order)
- Carrying (holding) cost - interest, storage, insurance, obsolescence, pilferage (15-30% of value per year)
- Shortage (stock-out) cost - lost sales, idle labour, goodwill - Purchase cost - price × quantity

8. Inventory Techniques

(a) Economic Order Quantity (EOQ) - Wilson's formula

The order size that gives the lowest total of ordering + carrying cost . At EOQ: ordering cost = carrying cost.



Formula

- EOQ = sqrt(2 x D x S ÷ H) where D = annual demand, S = ordering cost per order, H = carrying cost per unit per year (= i x unit price)
- Orders per year N = D ÷ EOQ ; Cycle time = 12 months (or 360 days) ÷ N ; Total cost = sqrt(2 x D x S x H)
- Assumptions: constant demand, constant lead time, no stock-outs, no quantity discount, instant replenishment

Example

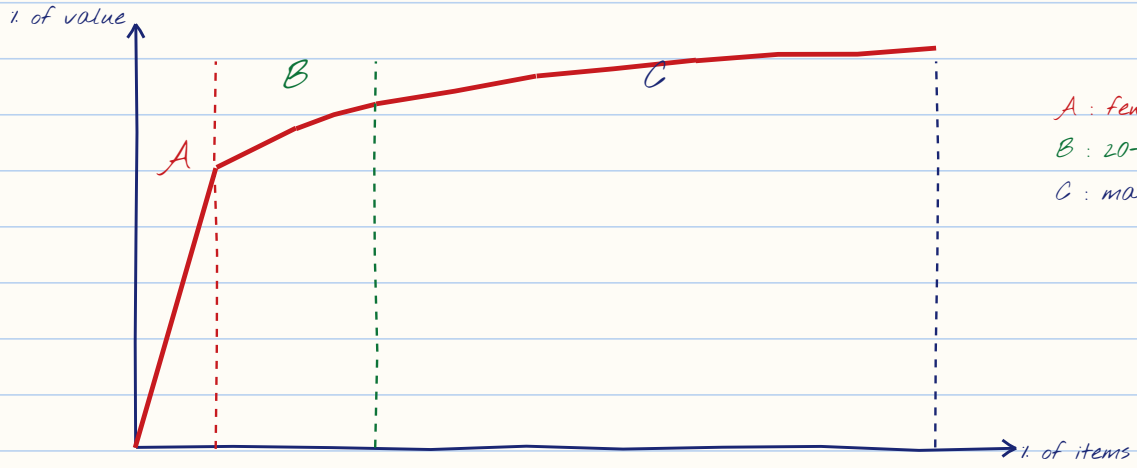
- D = 3600 units/year, S = Rs.200 per order, H = Rs.4 per unit/year
- EOQ = sqrt(2 x 3600 x 200 ÷ 4) = sqrt(360000) = 600 units
- Orders per year = 3600 ÷ 600 = 6 ; interval = 2 months
- Ordering cost = 6 x 200 = Rs.1200 ; Carrying cost = (600 ÷ 2) x 4 = Rs.1200 ; Total = Rs.2400 (equal costs - verified!)

Stock levels

- Reorder level (ROL) = maximum usage x maximum lead time (or = lead-time demand + safety stock)
- Minimum level = ROL - (normal usage x normal lead time) ; Maximum level = ROL + ROQ - (min usage x min lead time)
- Danger level = normal usage x emergency lead time ; Safety stock = buffer to cover uncertainty

(b) ABC Analysis (Always Better Control)

Based on Pareto's 80-20 rule - classify items by annual consumption value (annual usage x unit price) and control each class differently.



Class	% of items	% of value	Control policy
A	10-20%	70-80%	Tight control, accurate records, frequent review, small safety stock, careful vendor selection, EOQ
B	20-30%	15-20%	Moderate control, periodic review, reasonable records
C	50-70%	5-10%	Simple control, bulk ordering, large safety stock, minimum paperwork

- Steps: list items → annual usage value → arrange in descending order → compute cumulative % → divide into A, B, C
- Other classifications: VED (vital, essential, desirable), FSN (fast, slow, non-moving), HML (high, medium, low price), SDE (scarcity), JIT or Kanban (just-in-time)

B. MARKETING MANAGEMENT

9. Concept & Meaning

Marketing = human activity directed at satisfying needs and wants through exchange processes (Kotler). AMA : "Marketing is the process of planning and executing conception, pricing, promotion and distribution of ideas, goods, services to create exchanges that satisfy individual and organisational goals."

Era or concept	Focus	Slogan
Production	Availability, low cost	"Make more"
Product	Quality, features	"Better product sells"

Era or concept	Focus	Slogan
Selling	Aggressive promotion	"Sell what you make"
Marketing	Customer needs and profit	"Make what sells"
Societal	Customer + society welfare	"Customer, company, community"

Nature of marketing

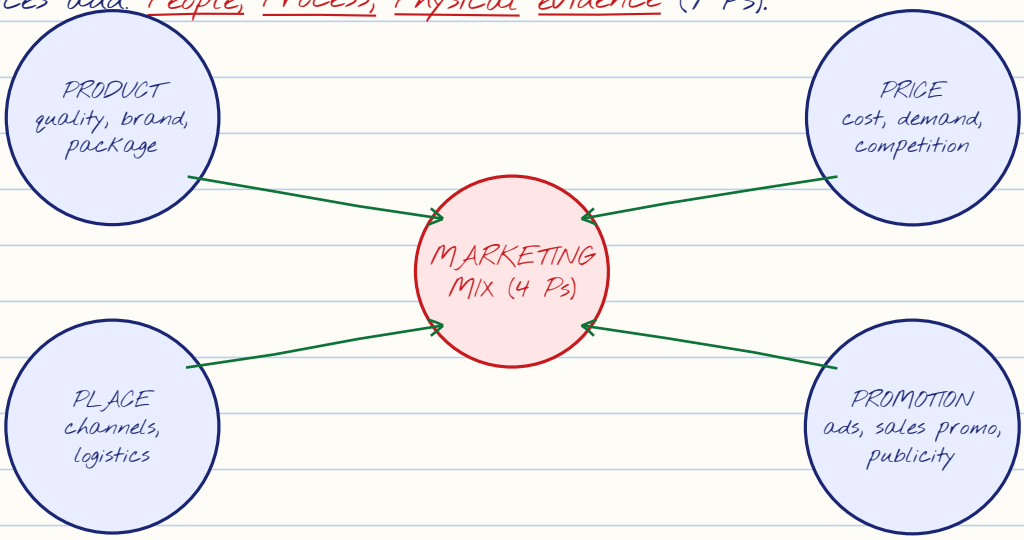
- Needs-oriented - Exchange process - Customer-centred - Both a science and an art - Dynamic and continuous - Creates 4 utilities: form, place, time, possession

Functions of marketing

- Exchange functions : buying, selling - Physical functions : transport, storage, packaging
- Facilitating functions : financing, risk-taking, grading and standardisation, market information
- Others : market research, product planning, pricing, promotion, distribution, after-sales service
- Difference: Marketing = need-based, whole process (before and after sale), Selling = product-based, ends with the sale

10. Marketing Mix

Neil Borden coined, E. Jerome McCarthy gave the 4 Ps. Marketing mix = controllable variables combined to get desired response from the target market. For services add: People, Process, Physical evidence (7 Ps).



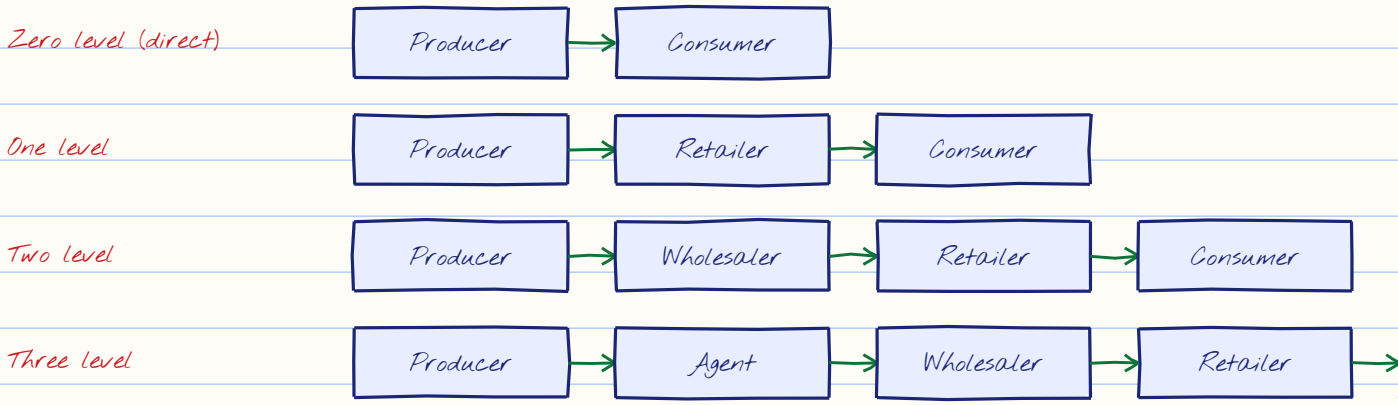
- PRODUCT : features, quality, design, brand, packaging, labelling, warranty,

product range, product life cycle

- PRICE : list price, discounts, credit terms. Methods: cost-plus, break-even, competition-based, value-based, penetration (low) and skimming (high) pricing
- PLACE : distribution channels, coverage, inventory, transport, warehouses, outlets
- PROMOTION : advertising, personal selling, sales promotion, publicity or PR, direct marketing
- Marketing strategies need consistency between the Ps (a luxury brand = premium product + high price + exclusive outlets + prestige ads)

II. Channels of Distribution

A channel = path or chain of intermediaries through which goods move from producer to consumer. Intermediaries: wholesalers, retailers, agents, brokers, distributors, C&F agents .



Factors for choosing a channel

- Product (perishable → short, technical → direct) - Market (size, geography, buyers' habits) - Company (finance, control wanted)
- Middlemen (availability, cost) - Competition - Cost or margin of channel - Legal and environmental factors

- Modern channels: e-commerce, franchising, direct selling, vertical marketing systems (VMS), omni-channel retail

12. Advertisement & Sales Promotion

Advertising

Advertising = paid, non-personal presentation of ideas, goods or services by an identified sponsor through mass media. Goals: inform, persuade, remind; build brand image.

- Media: Print (newspapers, magazines), Broadcast (TV, radio), Outdoor (hoardings, transit), Digital (social media, search, influencer, e-mail), cinema
- Merits: wide reach, creates demand, brand building, low cost per contact - Demerits: costly, no instant feedback, impersonal, may mislead (ASCI controls ads)
- Model - AIDA : Attention → Interest → Desire → Action

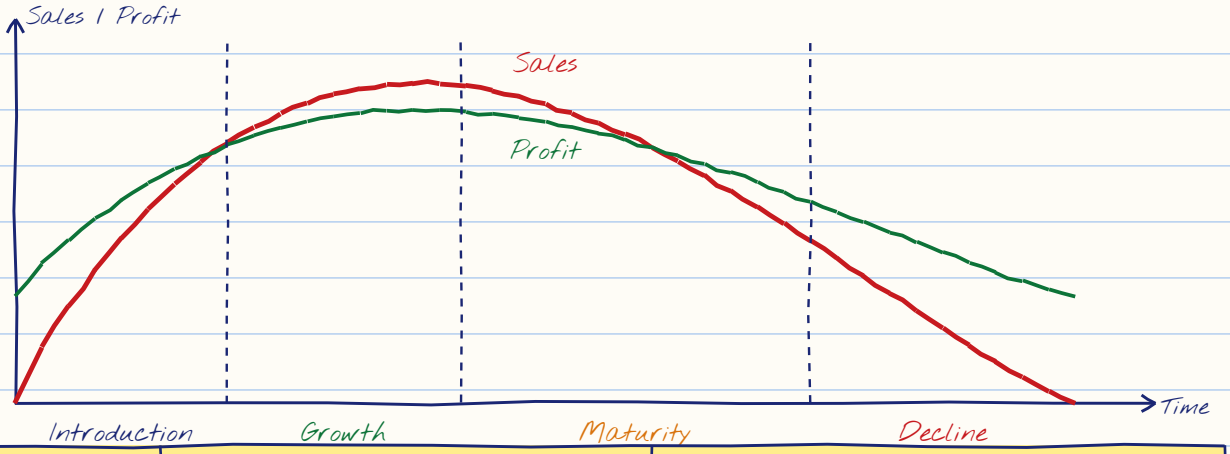
Sales promotion

- Consumer promotion : free samples, coupons, price-off, buy-1-get-1, contests, gifts, cashback, loyalty points, exchange offers
- Trade promotion : trade discounts, dealer contests, display allowances, trade shows
- Sales-force promotion : bonus, commission, sales contests, incentive trips
- Advertising = long-term, pull strategy ; Sales promotion = short-term, immediate sales ; Personal selling = face-to-face

Design an advertisement (practical steps)

- 1. Set objective and target audience 2. Fix budget 3. Create message (USP, headline, slogan) 4. Choose media 5. Design (visual, colour, layout, call-to-action) 6. Test and launch 7. Measure effect (recall, sales)
- Example ("Fresh Sip" juice): Headline "100% fruit, 0% worry" ; Visual: fruit splash ; Offer: buy 2 get 1 free this month ; CTA: scan QR to order ; Media: Instagram + college hoardings

13. Marketing Strategies by Product Life Cycle (PLC)



Stage	Features	Marketing strategy
Introduction	Low sales, high cost, losses, few competitors	Heavy promotion to create awareness, skimming or penetration price, limited distribution, trial offers
Growth	Rapid sales and profit rise, competitors enter	Improve quality, add features, build brand preference, widen distribution, reduce price a little
Maturity	Peak sales, slowing growth, intense competition	Product modification, market modification, new uses, discounts, defend share, loyalty programmes
Decline	Sales and profit fall, obsolescence	Harvest or divest, cut cost and promotion, drop weak products, serve loyal niche, or reinvent

14. Important Questions (Unit II)

1. Define plant layout. Explain its objectives, principles and types.
2. Compare job, batch and mass production.
3. What is work study? Explain method study and work measurement.
4. Explain SQC and the control charts (X-bar, R, p, c) with an example.
5. Define inventory. Explain its functions and types. Derive EOQ and solve a problem.
6. Explain ABC analysis with a neat graph.
7. Explain marketing mix. Discuss marketing strategies at different stages of

PLC.

8. Discuss channels of distribution and sales promotion methods.

Last-minute revision

- EOQ = sqrt(2DS ÷ H) ; ordering cost = carrying cost at EOQ ; ABC = 80-20 rule ; SREDDIM ; UCL = X-bar + A2R-bar ; 4Ps ; AIDA ; Intro-Growth-Maturity-Decline